

Problem Setup

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Problem Setup

We are given the quadratic equation

$$x^2 + 3x + 1 = 0$$

and we are asked to find all real (or complex) values of x that satisfy it.

Strategy and Key Principles

The standard method for solving a quadratic equation $ax^2 + bx + c = 0$ is the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This formula is derived from completing the square and works for any coefficients $a \neq 0$. We will apply it directly, computing the discriminant $b^2 - 4ac$ first.

Detailed Solution

$$a = 1,$$

$$b = 3,$$

$$c = 1,$$

$$\begin{aligned} \text{Discriminant } D &= b^2 - 4ac \\ &= 3^2 - 4 \cdot 1 \cdot 1 \\ &= 9 - 4 \\ &= 5, \end{aligned}$$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{D}}{2a} \\ &= \frac{-3 \pm \sqrt{5}}{2 \cdot 1} \\ &= \frac{-3 \pm \sqrt{5}}{2}. \end{aligned}$$

Final Answer

$$x = \frac{-3 \pm \sqrt{5}}{2}$$

Key Insights

- The discriminant $D = b^2 - 4ac$ determines the nature of the roots: $D > 0$ yields two distinct real roots, as seen here.
- A common mistake is to forget the factor $2a$ in the denominator of the quadratic formula; always verify that the denominator is $2a$, not just 2.
- The quadratic formula works for any quadratic, regardless of whether the coefficients are integers, fractions, or irrational numbers.